expense of their members, or at the expense of other groups. Rather, each gene is seen as pursuing its own self-interested agenda against the background of the other genes in the gene pool—the set of candidates for sexual shuffling within a species. Those other genes are part of the environment in which each gene survives, in the same way as the weather, predators and prey, supporting vegetation and soil bacteria are parts of the environment. From each gene's point of view, the 'background' genes are those with which it shares bodies in its journey down the generations. In the short term, that means the other members of the genome. In the long term, it means the other genes in the gene pool of the species. Natural selection therefore sees to it that gangs of mutually compatible—which is almost to say cooperating—genes are favoured in the presence of each other. At no time does this evolution of the 'cooperative gene' violate the fundamental principle of the selfish gene. Chapter 5 develops the idea, using the analogy of a rowing crew, and Chapter 13 takes it further.

Now, given that natural selection for selfish genes tends to favour cooperation among genes, it has to be admitted that there are some genes that do no such thing and work against the interests of the rest of the genome. Some authors have called them outlaw genes, others ultra-selfish genes, yet others just 'selfish genes'—misunderstanding the subtle difference from genes that cooperate in self-interested cartels. Examples of ultra-selfish genes are the meiotic drive genes described on pages 304–6, and the 'parasitic DNA' originally proposed on pages 56–7 and developed further by various authors under the catchphrase 'Selfish DNA'. The uncovering of new and ever more bizarre examples of ultra-selfish genes has become a feature of the years since this book was first published.*

*The Selfish Gene* has been criticized for anthropomorphic personification and this too needs an explanation, if not an apology. I employ two levels of personification: of genes, and of organisms. Personification of genes really ought not to be a problem, because no sane person thinks DNA molecules have conscious personalities, and no sensible reader would impute such a delusion to an author. I once had the honour of hearing the great molecular biologist Jacques Monod talking about creativity in science. I have forgotten his exact words, but he said approximately that, when trying to think through a chemical problem, he would ask himself what he would do if he were an electron. Peter Atkins,